

Middle East Technical University
Northern Cyprus Campus

Electrical and Electronics Engineering

EEE 212 Semiconductor Devices and Modeling

Homework

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Introduction

MOSFETs are the fundamental building blocks of digital logic circuits. In this workshop, the SiliWiz web based simulator is used to draw a simplified layout of an n-type MOSFET, layer by layer, and then simulate its electrical behavior. This helps visualize the role of each fabrication layer and the principles behind transistor switching.

The simulator demonstrates the following:

- Creation of diffusion regions, polysilicon gate, contacts, and metal interconnects
- Formation of the conductive channel based on applied voltages
- How logic gates are formed from transistors

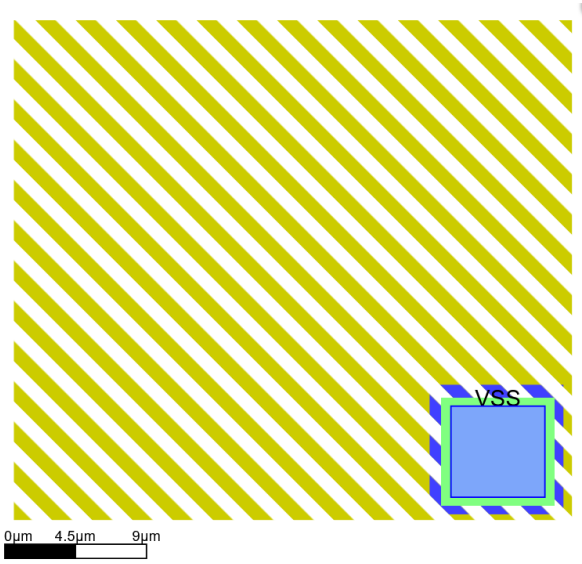
SiliWiz is a free, web based simulator that lets the user draw a transistor layout layer by layer, as in real fabrication, and then simulate its behavior under applied voltages. It allows an NMOS transistor to be built from diffusion areas, polysilicon, contacts, and metal wires, and to observe channel formation and switching when voltages are applied to the gate, source, and drain.

Objective

To design and simulate an NMOS transistor using SiliWiz and explore its switching behavior by building a layer by layer layout and measuring the drain current as a function of gate voltage.

Example 1: Basic Wafer and Substrate Bias

The base layer of a silicon chip is the **substrate**, usually a P-doped silicon wafer. A small amount of impurity is added to raise its conductivity, turning it from an insulator into a semiconductor. In SiliWiz the p-substrate is the large yellow-striped region covering the canvas.



(a) Layout view: p-substrate (yellow) with p tap and metal1 labeled `vss` in the bottom-right corner.



(b) Cross Section view: vertical stack of p-substrate, p tap, metal1 via, and metal1. DRC OK confirms no violations.

Figure 1: Example 1: p-substrate with VSS body bias connection via p tap.

As shown in Figure 1a, a **p tap** is placed in the bottom-right corner. Since the p-substrate body must stay at 0 V, it is connected to `vss` through the p tap, which forms an ohmic contact. On top of the p tap, a **metal1 via** and a **metal1** layer are stacked and labeled `vss`, giving the path:

$$\texttt{vss} \rightarrow \textbf{metal1} \rightarrow \textbf{metal1 via} \rightarrow \textbf{p tap} \rightarrow \textbf{p substrate}$$

Figure 1b shows the cross section through the p-tap region, with the layers visible from bottom to top: p-substrate, p tap, metal1 via, and metal1. The **DRC OK** status confirms the layout has no design rule violations.

Before Simulation: Screenshots of Design [30%]

Figure 2 shows the completed NMOS layout before simulation. It consists of a **p-substrate** body, a horizontal **n-diffusion** band forming the source and drain, a vertical **polysilicon** gate crossing the diffusion, and three metal1 contacts labeled `vss` (source), `vdd` (drain), and `in` (gate). The second `vss` in the bottom-right corner is the p-tap body bias from Example 1, which holds the substrate at 0 V.

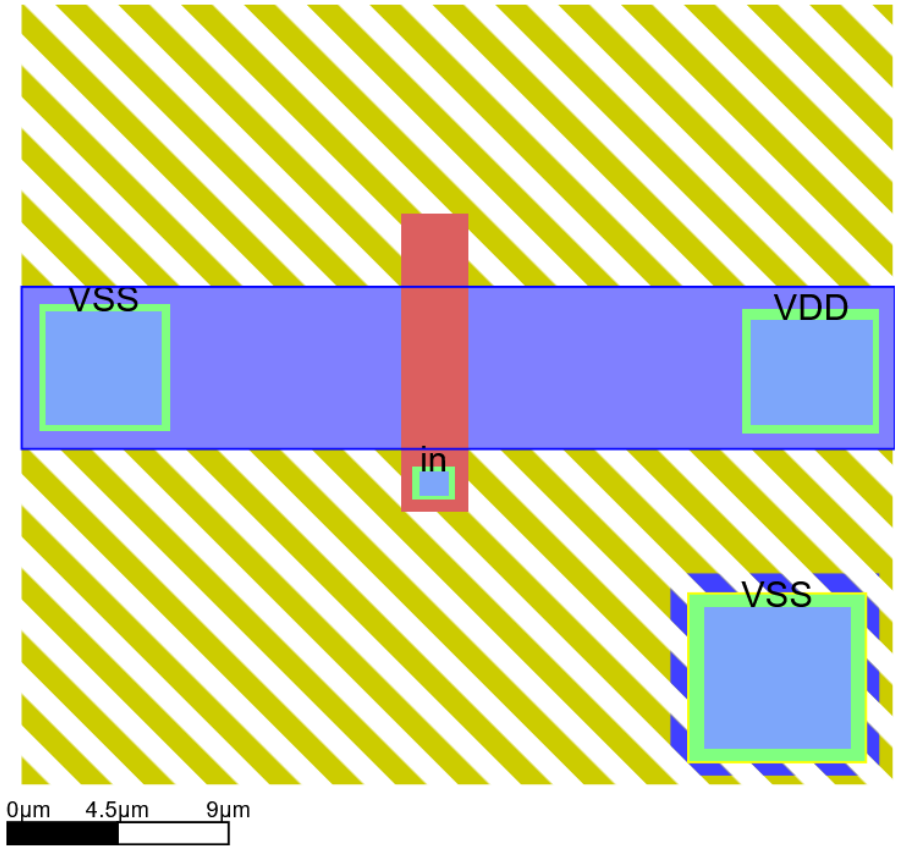
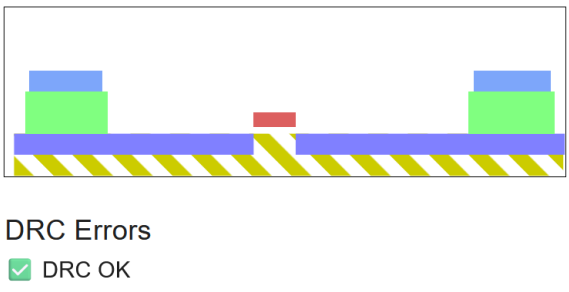
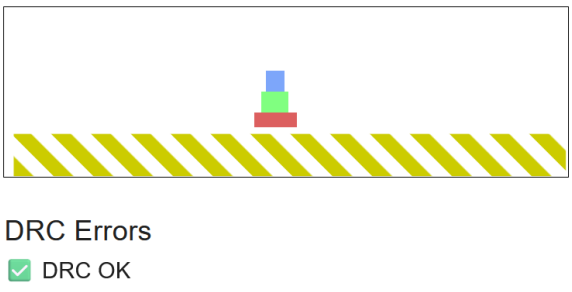


Figure 2: NMOS layout before simulation: p-substrate, n-diffusion (source and drain), polysilicon gate, metal1 contacts labeled `vss`, `vdd`, and `in`, and p-tap body bias. DRC OK.

Figures 3a and 3b show two cross section views at different slider positions. Figure 3a cuts through the transistor center and shows the split n-diffusion on either side of the gate, confirming that the gate divides the diffusion into source and drain. Figure 3b cuts through the gate contact and shows the polysilicon, metal1 via, and metal1 stack forming the `in` connection.



(a) Cross section through the transistor center: source (left), polysilicon gate (red, center), and drain (right) are visible. DRC OK.

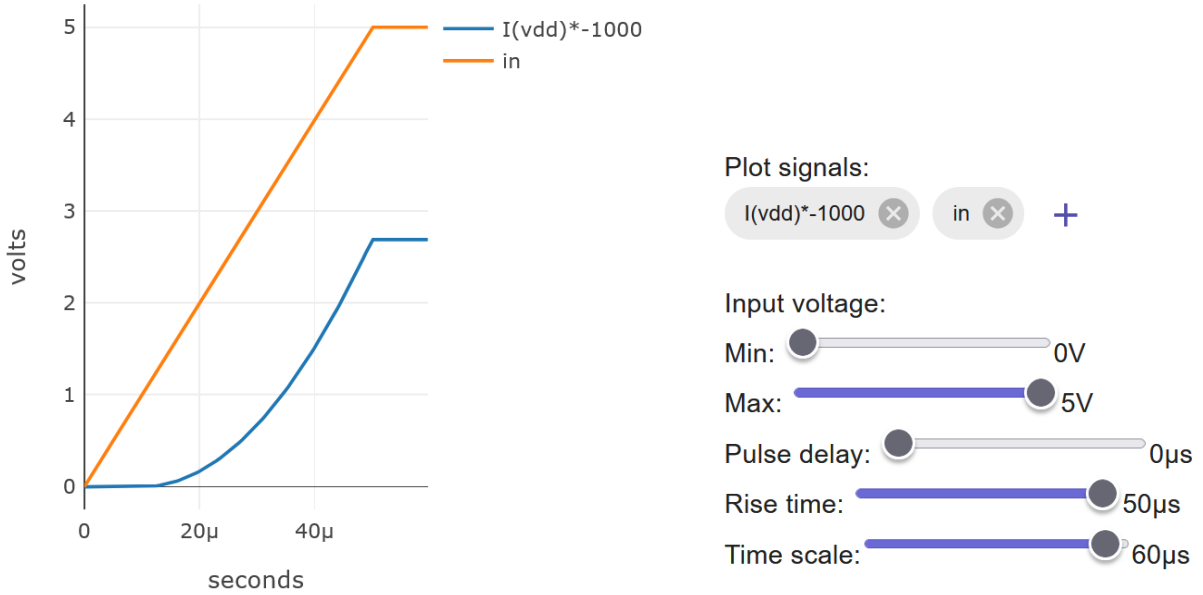


(b) Cross section through the gate contact: polysilicon (red), metal1 via (green), and metal1 (blue) stack forming the `in` terminal.

Figure 3: Cross section views of the NMOS layout at two slider positions.

After Simulation: Screenshots of Design [30%]

Figure 4 shows the simulation result. The gate voltage `in` (orange) is swept from 0 V to 5 V over a 50 μs rise time, and the drain current `I(vdd)*-1000` (blue) is plotted in milliamps.



(a) Simulation plot: gate voltage `in` (orange) and drain current `I(vdd)*-1000` (blue) in mA versus time.

(b) Simulation settings: input swept from 0 V to 5 V, rise time 50 μs, time scale 60 μs.

Figure 4: SiliWiz simulation result for the NMOS transistor.

The blue curve stays near zero until about $t = 20\,\mu$s, where the gate voltage rises enough above the threshold voltage V_{th} to form a channel beneath the gate. The drain current then rises sharply and levels off in saturation, reaching about 2.7 mA at $V_{GS} = 5$ V.

Observations [40%]

1. Simulation Observations

The simulation shows three phases as the gate voltage `in` is swept from 0 V to 5 V.

Off state. Below V_{th} (up to about $t = 10\,\mu$s) no channel forms, the p-type region between the two n-diffusion areas stays intact, and the drain current is zero. The transistor acts as an open circuit.

Turn on. Around $t \approx 10$ to $15\,\mu$s the gate voltage crosses V_{th}. The gate field pulls electrons to the surface beneath the polysilicon, forming an n-type channel that links source and drain. Current starts to flow and the blue curve rises sharply.

Saturation. As V_{GS} approaches 5 V the current levels off near 2.7 mA. The channel is pinched off at the drain end and the current is set mainly by V_{GS}, so the transistor acts as a voltage controlled current source.

Sign convention. The simulator treats current into `vdd` as negative, so multiplying by -1000 gives a positive milliamp value for plotting.

1.1 Effect of Increasing Polysilicon Size

The polysilicon gate sets two parameters: the **channel length** L (source to drain direction) and the **channel width** W (perpendicular direction).

Increasing channel length L. The saturation drain current is:

$$I_D = \frac{1}{2}\,\mu_n\,C_{ox}\,\frac{W}{L}\,(V_{GS} - V_{th})^2$$

A larger L lowers W/L and reduces I_D, since electrons must drift further under the same field. The gate capacitance also grows:

$$C_G = C_{ox} \cdot W \cdot L$$

which slows switching. The transconductance drops as well:

$$g_m = \mu_n\,C_{ox}\,\frac{W}{L}\,(V_{GS} - V_{th})$$

so the gate controls the current less effectively. A short gate therefore gives the most saturation current.

Increasing channel width W. A wider gate raises W/L, increasing I_D and g_m. This is why the n-diffusion here spans the full width: a larger W gives more current, matching our 2.7 mA result.

Fabrication: Bonus [10%]

Reference: IHP cleanroom tour video, "Inside the cleanroom: How computer chips get made!" https://youtu.be/aBDJQ9NYTEU

1. What is Photolithography?

Photolithography is the technique used to print millions of MOSFET gates onto a wafer with a single flash of light. First, a light sensitive photoresist is spread evenly over the

wafer by spin coating. The wafer is then loaded with a photomask, which at IHP is a quartz plate with a patterned chromium layer protected by a thin foil that keeps dust away. Light passes through the open parts of the mask and a lens projects the pattern onto the wafer at about four times smaller. IHP uses two tools: an i-line stepper at 365 nm for larger structures such as the top metal layers, and a deep ultraviolet (DUV) stepper at 248 nm for the smaller, critical layers. After exposure, the illuminated resist changes chemically and is dissolved during development, leaving a pattern that defines where the next step acts.

In addition to what is shown in the video, the most advanced photolithography technology today is **High NA EUV** (High Numerical Aperture Extreme Ultraviolet lithography). It is not used at IHP and does not appear in the video; IHP's tools are several generations behind the leading edge.

Resolution follows from the Rayleigh equation for the critical dimension (CD), the smallest printable feature:

$$\mathrm{CD} = k_1 \cdot \frac{\lambda}{\mathrm{NA}}$$

where λ is the wavelength, NA the numerical aperture, and k_1 a process constant. A smaller CD comes from a shorter λ or a larger NA. Wavelength is the main factor. IHP's 365 nm i-line is in the near ultraviolet (UV-A), close to visible violet, and its 248 nm DUV is in the deep ultraviolet (UV-C). EUV goes much further at 13.5 nm, in the extreme ultraviolet band just above the soft X-ray region. This very short wavelength is what lets EUV draw features that longer wavelengths cannot. As an analogy, a pen with a 0.7 mm tip cannot draw lines as fine as one with a 0.05 mm tip: IHP draws with the thick tip of its i-line and DUV tools, while High NA EUV uses an ultra-fine tip. High NA EUV also raises NA from 0.33 to 0.55, adding further resolution.

This higher resolution lets High NA EUV print the finest patterns in a single exposure, while DUV must split one pattern across several masks and exposures, a slow and costly method called multi-patterning. Single-exposure patterning simplifies the process, reduces defects, lowers cost, and enables the sub-10 nm geometries used in the most advanced logic and memory chips. I first learned about EUV and High NA EUV from the Veritasium video "The World's Most Important Machine" (https://www.youtube.com/watch?v=MiUHjLxm3V0).

2. Why is masking important?

Masking lets a process step such as etching, doping, or deposition, which would otherwise affect the whole wafer, act only on selected areas. Each mask holds the pattern for one layer, and a single reticle can carry several layer patterns. Even one stray particle on the chromium can distort the projected image and cause defects, such as short circuits or broken metal lines, on every wafer exposed, which is why the mask is kept under a protective foil. Since the mask defines where each step acts, including where trenches are etched and where resist is opened for implantation, its accuracy decides whether the transistors and interconnects work.

3. Why is ion implantation used in manufacturing?

After the resist mask is developed, the wafer goes into an ion implantation tool. The wafers sit on a rotating wheel while dopant atoms, usually phosphorus, arsenic, or boron, are accelerated into a beam and driven into the exposed silicon. The tool separates ions by mass and sets their energy, giving independent control of the implant depth and concentration, which ranges from about one dopant per hundred silicon atoms to one per million. This control is why implantation is preferred: different regions can get different dopant types, depths, and densities just by changing the mask and settings. Because the implant damages the crystal, the wafer then undergoes rapid thermal processing, a short high temperature anneal that repairs the silicon and activates the dopants so they provide free carriers.